MILITARY SPECIFICATION

Monitor, Radio Frequency

ID-902A/AR

This specification has been approved
by the Bureau of Naval Weapons
Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide means for measurement of forward and reverse r-f power and VSWR of a transmitter and shall provide a visual ON/OFF indication for associated equipment.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

FSC 6625

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STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

Federal

FED-STD-595 Colors

PUBLICATIONS

ANA Bulletin No. 157 - Colors; List of Standard Aircraft Camoflage

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern.

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.2 Design and Construction:

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3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 2 pounds.

3.2.3 Total Operating Life - The equipment shall have minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.5 Control Panels - The equipment shall conform to the applicable requirements of Specification MIL-C-6781 except as specified herein. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.5.1 Finish - The surface finish materials shall be white and gray vinyl (number KDA2905 Bakelite Corporation or equivalent). The gray color shall be ANA Bulletin No. 157, number 621 Dark Gull Gray or FED-STD-595 number 36231 Gray.

3.2.5.1.1 Opaque Surface - The front face and edges of the plastic plate shall be coated with an initial white opaque coating and a second gray opaque coating to give a non-reflecting surface. The back face shall be coated with gray vinyl paint.

3.2.5.2 Color - When the lighting units are energized, the lettering, numbering and markings shall be white. The background shall be gray.

3.2.5.3 Contrast - The value of contrast when measured under daylight illumination shall be not less than 1.75.

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3.2.5.4 Illumination - The light assemblies shall be so located that all illuminated markings on the plastic lighting plate and components illuminated by the plastic lighting plate shall have a relatively uniform brightness over the range of lamp voltage 2 to 5.0 volts. When the lamps are operated at rated candlepower without a color filter, the dimmest markings shall be not less than 0.5 foot-lamberts and brightest markings shall be not more than 3.5 foot-lamberts.

3.2.5.5 Mounting - Gray head screws shall be employed to secure the plastic lighting plate to the mounting plate.

3.2.5.6 Control Knobs - Control knobs shall be gray in agreement with panel color.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ±0.5 VDC (Indicator)
	5.0 ±0.1 VDC (Panel Lamps)

3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environemental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of Specification MIL-E-5400. The vibration requirements of Curve I of MIL-E-5400 shall not apply to this equipment.

3.2.10 Warmup Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard or extreme service conditions.

3.2.11 Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 except as specified below and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power shall not exceed the specified amounts.

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(1) D-C Power, 28-V, Category 'B', 0.1 Amp.
(2) D-C Power, 5V, (Panel Lighting) 0.8 Amp.

3.2.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.2.12 Terminations - The following terminations shall be included in the input connector of the r-f monitor:

(1) Incident power input - Terminal for application of voltage developed by the directional coupler which is proportional to incident r-f power applied to an antenna or other suitable load from the associated transmitter.

(2) Reflected power input - Terminal for application of voltage proportional to r-f power reflected from the associated transmitter load.

(3) Flag indicator control -Terminal for ON/OFF control wire from associated equipment to energize the indicator simultaneously with the associated equipment.

(4) Indicator power input - Terminal for application of power required to actuate the indicator.

(5) Panel lamp power input - Terminal for application of power to panel lamps.

(6) Ground terminal - Ground return for all terminations.

3.2.13 Input Impedance - The d-c input impedance at the incident power input terminals shall be 1425 ohms ± 10 percent with the meter switch set at the INC PWR position and the PWR CAL control adjusted fully clockwise, and between 21.3K ohms and 31. 6K ohms when adjusted fully counterclockwise. The d-c input impedance at the reflected power input terminals shall be 1425 ohms ± 10 percent with the meter switch set at the VSWR position and the VSWR CAL control adjusted fully clockwise, and between 21.3K ohms and 31.6K ohms when adjusted fully counterclockwise.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.3.2 R-F Power and VSWR Indicator - Forward and reflected power and VSWR developed by the associated transmitter shall be visually displayed on a meter located on the front panel of the R-F Monitor. The meter shall include a dual scale graduated from 0-to 40-watts full scale and 1-to ∞-VSWR full scale.

3.3.2.1 R-F Power Indication - With the selector switch set to measure incident power and the calibrating control adjusting fully clockwise, the following values of current (within ±5.0 microamps) into the incident power terminals shall produce the power readings shown below under standard conditions.

Input Current u A	Power Reading (watts)
0	0
3.3	1
12.3	5
20.3	10
26.3	15
32.5	20
41.8	30
50.0	40

3.3.2.2 VSWR Indication - With the selector switch set to the VSWR CAL position and the calibrating control adjusted fully clockwise, the following values of current (within ±5.0 microamp) into the incident power terminals shall produce the VSWR readings shown below:

Input Current u A	VSWR Reading
0	1
11	2
20.3	3
29.6	5
38.6	10
43.9	20
50.0	∞

3.3 Flag Indicator - The power ON/OFF status of the associated equipment shall be visually displayed on the front panel of the r-f monitor by means of a suitable flag type indicator. The letters OFF shall be normally displayed on the indicator face. When connected to a suitable d-c power source and the indicator control lead is grounded, the letters ON shall appear on the indicator face. The OFF indication shall reappear upon removal of the ground from the control lead.

3.4 Detail Requirements -

3.4.1 Monitor, Radio Frequency ID-902A/AR - The r-f monitor shall meet the following requirements:

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3.4.1.1 Function - The r-f monitor shall provide means to measure incident r-f power, reflected r-f power and VSWR of a UHF transmitter when properly connected to a suitable directional coupler associated with the transmitter. It shall also provide a visual indication of the ON/OFF status of associated equipment.

3.4.1.2 Form Factor - The over-all dimensions of the r-f monitor shall not exceed 5-3/4 inches in width, 3-3/4 inches in length, and 3 - 1/2 inches in depth.

3.4.1.3 Weight - The weight of the r-f monitor shall not exceed 2.0 pounds.

3.4.1.4 Contents - The r-f monitor shall contain an r-f power and VSWR indicating meter with associated switching and calibrating circuitry which shall permit measurement of r-f power output and VSWR of a UHF transmitter from voltage developed by a suitable directional coupler contained within the transmitter. The meter circuit shall include a 60 uf capacitance to maximize meter readings when measuring pulsed r-f power. The unit also shall include a flag-type indicator to provide a visual indication of the ON/OFF status of associated equipment of the communication system.

3.4.1.5 Controls - The following controls shall be located on the front panel of the r-f monitor:

(1) PWR CAL - Screwdriver adjustment to calibrate forward power meter reading.

(2) VSWR CAL - Knob adjustment to set meter reading for VSWR measurement.

(3) Four-position meter switch with knob adjustment to select INC PWR, REFL PWR, VSWR CAL and VSWR measurement functions.

3.4.1.6 Electrical Connections - Connections to Power Input shall be provided by a Bendix PT02A-12-10P(or an approved equivalent) receptacle designated 'J2901'.

3.4.2 Associated Equipment - The r-f monitor shall operate with the following associated equipment which shall not be supplied as part of this equipment.

(1)	T-713/ART-36	UHF Transmitter
(2)	AN/ASW-14A	Digital Data Comm. Control Set

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Test

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment, submitted for service approval tests, the contractor also shall submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has passed all individual tests successfully. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under contract shall cease until proper corrective action is approved and accomplished. Corrective action also shall be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

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(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering change.

(2) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test Procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected

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to this test. (Equipments which have passed the sampling tests or special tests successfully may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
D-C voltage	27.0±2.0 volts (Indicator)
	5.0±0.1 volts (Panel lamps)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

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4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows.

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been tested and inspected previously by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and Limiting values of performance data shall be determined at input voltage of 27.5±0.5 and 5.0 V ±0.1 volts d-c. These data are to be used in testing the equipment at installation points for compliance with the minimum acceptable standard of performance.

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6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and abbrev. title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY (Print name and title)	DATE

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